Java String Methods — Complete Practical Guide

Important java.lang.String methods with syntax, purpose, examples, output, and interview notes. Examples use Java 8+ unless a method is marked with its minimum Java version.

1. Core String Methods

length()

Syntax: `int length()`

What it does: Returns the number of characters.

Example:

```
String s = "Java";
System.out.println(s.length());
```

Output: 4

isEmpty()

Syntax: `boolean isEmpty()`

What it does: Returns true when the String length is 0.

Example:

```
System.out.println("".isEmpty());
System.out.println("Java".isEmpty());
```

Output: true
false

isBlank()

Syntax: `boolean isBlank()`

What it does: Returns true when the String is empty or contains only Unicode whitespace. Java 11+.

Example:

```
System.out.println(" ".isBlank());
System.out.println("Java".isBlank());
```

Output: true
false

charAt()

Syntax: `char charAt(int index)`

What it does: Returns the character at the specified zero-based index.

Example:

```
System.out.println("Java".charAt(2));
```

Output: v

codePointAt()

Syntax: `int codePointAt(int index)`

What it does: Returns the Unicode code point beginning at the specified index.

Example:

```
System.out.println("A".codePointAt(0));
```

Output: 65

codePointBefore()

Syntax: `int codePointBefore(int index)`

What it does: Returns the Unicode code point before the specified index.

Example:

```
System.out.println("ABC".codePointBefore(2));
```

Output: 66

codePointCount()

Syntax: `int codePointCount(int beginIndex, int endIndex)`

What it does: Counts Unicode code points in the specified range.

Example:

```
System.out.println("ABC".codePointCount(0, 3));
```

Output: 3

offsetByCodePoints()

Syntax: `int offsetByCodePoints(int index, int codePointOffset)`

What it does: Returns an index offset by a specified number of Unicode code points.

Example:

```
System.out.println("ABC".offsetByCodePoints(0, 2));
```

Output: 2

getChars()

Syntax: `void getChars(int srcBegin, int srcEnd, char[] dst, int dstBegin)`

What it does: Copies characters from this String into a destination char array.

Example:

```
char[] a = new char[4];
"Java".getChars(0, 4, a, 0);
System.out.println(a);
```

Output: Java

getBytes()

Syntax: `byte[] getBytes()`

What it does: Encodes the String using the platform default charset.

Example:

```
byte[] b = "ABC".getBytes();
System.out.println(b.length);
```

Output: 3

getBytes(Charset)

Syntax: `byte[] getBytes(Charset charset)`

What it does: Encodes using a specified charset; explicit charset is preferred.

Example:

```
byte[] b = "ABC".getBytes(java.nio.charset.StandardCharsets.UTF_8);
System.out.println(b.length);
```

Output: 3

equals()

Syntax: `boolean equals(Object anObject)`

What it does: Checks exact, case-sensitive String content equality.

Example:

```
System.out.println("Java".equals("Java"));
System.out.println("Java".equals("java"));
```

Output: true
false

equalsIgnoreCase()

Syntax: `boolean equalsIgnoreCase(String anotherString)`

What it does: Compares String content while ignoring case.

Example:

```
System.out.println("Java".equalsIgnoreCase("JAVA"));
```

Output: true

contentEquals()

Syntax: `boolean contentEquals(CharSequence cs)`

What it does: Compares this String with another CharSequence.

Example:

```
System.out.println("Java".contentEquals(new StringBuilder("Java")));
```

Output: true

compareTo()

Syntax: `int compareTo(String anotherString)`

What it does: Lexicographically compares two Strings; returns 0 when equal.

Example:

```
System.out.println("apple".compareTo("banana"));
```

Output: negative value

compareToIgnoreCase()

Syntax: `int compareToIgnoreCase(String str)`

What it does: Lexicographically compares while ignoring case.

Example:

```
System.out.println("java".compareToIgnoreCase("JAVA"));
```

Output: 0

startsWith()

Syntax: `boolean startsWith(String prefix)`

What it does: Checks whether the String begins with the specified prefix.

Example:

```
System.out.println("Java Programming".startsWith("Java"));
```

Output: true

startsWith(prefix, offset)

Syntax: `boolean startsWith(String prefix, int toffset)`

What it does: Checks a prefix beginning at a specified index.

Example:

```
System.out.println("Java Programming".startsWith("Programming", 5));
```

Output: true

endsWith()

Syntax: `boolean endsWith(String suffix)`

What it does: Checks whether the String ends with the specified suffix.

Example:

```
System.out.println("Java.java".endsWith(".java"));
```

Output: true

contains()

Syntax: `boolean contains(CharSequence s)`

What it does: Checks whether the String contains the specified character sequence.

Example:

```
System.out.println("Java Programming".contains("gram"));
```

Output: true

indexOf(String)

Syntax: `int indexOf(String str)`

What it does: Returns the first index of a substring, or -1 if not found.

Example:

```
System.out.println("banana".indexOf("na"));
```

Output: 2

indexOf(int)

Syntax: `int indexOf(int ch)`

What it does: Returns the first index of a character/code point, or -1.

Example:

```
System.out.println("banana".indexOf('a'));
```

Output: 1

indexOf(String, fromIndex)

Syntax: `int indexOf(String str, int fromIndex)`

What it does: Searches for a substring starting at or after the specified index.

Example:

```
System.out.println("banana".indexOf("na", 3));
```

Output: 4

lastIndexOf(String)

Syntax: `int lastIndexOf(String str)`

What it does: Returns the last index of a substring, or -1.

Example:

```
System.out.println("banana".lastIndexOf("na"));
```

Output: 4

lastIndexOf(int)

Syntax: `int lastIndexOf(int ch)`

What it does: Returns the last index of a character/code point.

Example:

```
System.out.println("banana".lastIndexOf('a'));
```

Output: 5

substring(begin)

Syntax: `String substring(int beginIndex)`

What it does: Returns characters from beginIndex through the end.

Example:

```
System.out.println("JavaProgramming".substring(4));
```

Output: Programming

substring(begin, end)

Syntax: `String substring(int beginIndex, int endIndex)`

What it does: Returns the range [beginIndex, endIndex); endIndex is excluded.

Example:

```
System.out.println("JavaProgramming".substring(0, 4));
```

Output: Java

subSequence()

Syntax: `CharSequence subSequence(int beginIndex, int endIndex)`

What it does: Returns a CharSequence for the specified range.

Example:

```
System.out.println("JavaProgramming".subSequence(4, 11));
```

Output: Program

concat()

Syntax: `String concat(String str)`

What it does: Appends another String and returns the resulting String.

Example:

```
System.out.println("Hello".concat(" World"));
```

Output: Hello World

replace(char,char)

Syntax: `String replace(char oldChar, char newChar)`

What it does: Replaces every occurrence of one character.

Example:

```
System.out.println("banana".replace('a', 'o'));
```

Output: bonono

replace(CharSequence,CharSequence)

Syntax: `String replace(CharSequence target, CharSequence replacement)`

What it does: Replaces every literal occurrence of a character sequence.

Example:

```
System.out.println("Java Java".replace("Java", "Python"));
```

Output: Python Python

replaceFirst()

Syntax: `String replaceFirst(String regex, String replacement)`

What it does: Replaces the first substring matching a regular expression.

Example:

```
System.out.println("a1 b2".replaceFirst("\\d", "X"));
```

Output: aX b2

replaceAll()

Syntax: `String replaceAll(String regex, String replacement)`

What it does: Replaces every substring matching a regular expression.

Example:

```
System.out.println("a1 b2".replaceAll("\\d", "X"));
```

Output: aX bX

matches()

Syntax: `boolean matches(String regex)`

What it does: Checks whether the entire String matches a regular expression.

Example:

```
System.out.println("12345".matches("\\d+"));
```

Output: true

split(regex)

Syntax: `String[] split(String regex)`

What it does: Splits the String around matches of a regular expression.

Example:

```
String[] a = "Java,Python,C".split(",");
System.out.println(java.util.Arrays.toString(a));
```

Output: [Java, Python, C]

split(regex,limit)

Syntax: `String[] split(String regex, int limit)`

What it does: Splits using a regular expression with a maximum result-array length.

Example:

```
System.out.println(java.util.Arrays.toString("a-b-c".split("-", 2)));
```

Output: [a, b-c]

toLowerCase()

Syntax: `String toLowerCase()`

What it does: Converts the String to lowercase using the default locale.

Example:

```
System.out.println("JaVa".toLowerCase());
```

Output: java

toUpperCase()

Syntax: `String toUpperCase()`

What it does: Converts the String to uppercase using the default locale.

Example:

```
System.out.println("JaVa".toUpperCase());
```

Output: JAVA

trim()

Syntax: `String trim()`

What it does: Removes leading/trailing characters whose code points are <= U+0020.

Example:

```
System.out.println(" Java ".trim());
```

Output: Java

strip()

Syntax: `String strip()`

What it does: Removes leading/trailing Unicode whitespace. Java 11+.

Example:

```
System.out.println(" Java ".strip());
```

Output: Java

stripLeading()

Syntax: `String stripLeading()`

What it does: Removes leading Unicode whitespace. Java 11+.

Example:

```
System.out.println(" Java".stripLeading());
```

Output: Java

stripTrailing()

Syntax: `String stripTrailing()`

What it does: Removes trailing Unicode whitespace. Java 11+.

Example:

```
System.out.println("Java ".stripTrailing());
```

Output: Java

toCharArray()

Syntax: `char[] toCharArray()`

What it does: Converts the String into a new char array.

Example:

```
char[] a = "Java".toCharArray();
System.out.println(a[1]);
```

Output: a

valueOf()

Syntax: `static String valueOf(...)`

What it does: Converts primitives or objects into String form; several overloads exist.

Example:

```
String s = String.valueOf(123);
System.out.println(s + 1);
```

Output: 1231

copyValueOf()

Syntax: `static String copyValueOf(char[] data)`

What it does: Creates a String from a char array.

Example:

```
char[] a = {'J','a','v','a'};
System.out.println(String.copyValueOf(a));
```

Output: Java

format()

Syntax: `static String format(String format, Object... args)`

What it does: Creates a formatted String.

Example:

```
String s = String.format("Name: %s, Age: %d", "Ravi", 20);
System.out.println(s);
```

Output: Name: Ravi, Age: 20

join()

Syntax: `static String join(CharSequence delimiter, CharSequence... elements)`

What it does: Joins multiple elements using a delimiter. Java 8+.

Example:

```
System.out.println(String.join("-", "2026", "08", "19"));
```

Output: 2026-08-19

intern()

Syntax: `String intern()`

What it does: Returns the canonical representation from the String pool.

Example:

```
String a = new String("Java").intern();
String b = "Java";
System.out.println(a == b);
```

Output: true

repeat()

Syntax: `String repeat(int count)`

What it does: Repeats the String count times. Java 11+.

Example:

```
System.out.println("Hi ".repeat(3));
```

Output: Hi Hi Hi

lines()

Syntax: `Stream<String> lines()`

What it does: Returns a stream of lines separated by line terminators. Java 11+.

Example:

```
long n = "A\nB\nC".lines().count();
System.out.println(n);
```

Output: 3

indent()

Syntax: `String indent(int n)`

What it does: Adjusts indentation by n spaces; negative values can remove indentation. Java 12+.

Example:

```
System.out.println("A\nB".indent(2));
```

Output: A
B

transform()

Syntax: `<R> R transform(Function<? super String,? extends R> f)`

What it does: Applies a function to this String and returns the function result. Java 12+.

Example:

```
int n = "Java".transform(String::length);
System.out.println(n);
```

Output: 4

stripIndent()

Syntax: `String stripIndent()`

What it does: Removes incidental indentation from each line. Java 15+.

Example:

```
String s = " A\n B";
System.out.println(s.stripIndent());
```

Output: A
B

translateEscapes()

Syntax: `String translateEscapes()`

What it does: Converts escape sequences such as \\n and \\t. Java 15+.

Example:

```
System.out.println("A\\nB".translateEscapes());
```

Output: A
B

formatted()

Syntax: `String formatted(Object... args)`

What it does: Formats this String as a format string. Java 15+.

Example:

```
System.out.println("Name: %s".formatted("Ravi"));
```

Output: Name: Ravi

chars()

Syntax: `IntStream chars()`

What it does: Returns an IntStream of UTF-16 char values.

Example:

```
System.out.println("ABC".chars().sum());
```

Output: 198

codePoints()

Syntax: `IntStream codePoints()`

What it does: Returns an IntStream of Unicode code points.

Example:

```
System.out.println("ABC".codePoints().count());
```

Output: 3

2. Most Important Methods for Coding Interviews

Task	Methods
Length / empty checks	length(), isEmpty(), isBlank()
Character access	charAt()
Search	indexOf(), lastIndexOf(), contains()
Extract	substring(), subSequence()
Compare	equals(), equalsIgnoreCase(), compareTo()
Prefix / suffix	startsWith(), endsWith()
Replace	replace(), replaceFirst(), replaceAll()
Split / join	split(), String.join()
Case	toLowerCase(), toUpperCase()
Whitespace	trim(), strip(), stripLeading(), stripTrailing()
Convert	toCharArray(), getBytes(), valueOf()
Regex	matches(), split(), replaceFirst(), replaceAll()
Modern Java	repeat(), lines(), formatted(), strip(), isBlank()

3. Key Interview Notes

• String is immutable: methods such as concat(), replace(), substring(), toUpperCase(), etc. return a result rather than changing the original String.

• == compares references; equals() compares String contents.

• substring(beginIndex, endIndex) uses an exclusive end index.

• indexOf()/lastIndexOf() return -1 when the requested character or substring is not found.

• replace() performs literal replacement; replaceAll()/replaceFirst() interpret their first argument as a regular expression.

• trim() has legacy whitespace behavior; strip() is Unicode-aware and was introduced in Java 11.

• isEmpty() checks length == 0; isBlank() also treats whitespace-only strings as blank.

• StringBuilder and StringBuffer are separate mutable classes; their methods are not String methods.

4. Version Guide

Java version	Useful String additions
Java 8	join(), chars(), codePoints(), etc.
Java 11	isBlank(), lines(), strip(), stripLeading(), stripTrailing(), repeat()
Java 12	indent(), transform()
Java 15	formatted(), stripIndent(), translateEscapes()